

I. Discrete Distributions

A. Binomial Distribution

A discrete random variable X corresponding to an experiment done n times.

Concept	Formula
Probability Mass Function (PMF)	$P(X = r) = {}^nC_r p^r q^{n-r}$ where $r = 0, 1, \dots, n$
Mean (Expected Value)	$E[X] = np$
Variance	$\sigma^2 = Var[X] = npq$
Probability of Success	p
Probability of Failure	$q = 1 - p$

B. Poisson Distribution

A discrete random variable X follows Poisson's distribution if it has the following PMF. Used when n is very large and p is very low.

Concept	Formula
Probability Mass Function (PMF)	$P[X = r] = \frac{e^{-\lambda} \lambda^r}{r!}$ where $r = 0, 1, 2, \dots, n$
Mean and Variance	Mean = Variance = λ
Lambda (λ)	$\lambda = np$

C. Geometric Distribution

A discrete random variable taking non-negative values follows Geometric distribution if it has the following PMF.

Concept	Formula
Probability Mass Function (PMF)	$P[X = r] = q^{r-1} \cdot p$ (probability of 1st success on r -th trial)
Mean (Expected Value)	$E[X] = \frac{1}{p}$
Variance	Variance = $\frac{q}{p^2}$

II. Continuous Distributions

A. Normal Distribution

A continuous random variable X follows Normal Distribution with parameters μ and σ .

	Formula	Citation
Standardization (Z-score)	$Z = \frac{X - \mu}{\sigma}$	
Normal PDF	$\Phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ (where $\mu = 0, \sigma = 1$)	
Standard deviation (σ of sample mean) (Central Limit Theorem)	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ (population S.D. divided by $\sqrt{n}$)	

B. Exponential Distribution

A continuous random variable X , taking positive values, follows Exponential distribution with parameter θ .

Concept	Formula	Citation
Probability Density Function (PDF)	$f(x) = \theta e^{-\theta x}, \text{ for } x > 0$	
Mean	$\text{Mean} = \frac{1}{\theta}$	
Variance	$\text{Variance} = \frac{1}{\theta^2}$	

III. Correlation and Regression

A. Karl Pearson's Coefficient of Correlation

Measures the strength and direction of the linear relationship between X and Y .

Concept	Formula
Correlation Coefficient (r) (Deviation from Mean)	$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{\Sigma(x - \bar{x})^2 \cdot \Sigma(y - \bar{y})^2}}$
Alternative Form for r (Raw Data)	$r = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{(\Sigma x^2 - \frac{(\Sigma x)^2}{n})(\Sigma y^2 - \frac{(\Sigma y)^2}{n})}}$

B. Regression Lines and Coefficients

Concept	Formula
Regression Coefficient (b_{yx}) of Y on X	$b_{yx} = \frac{r\sigma_y}{\sigma_x}$
Regression Line of Y on X	$y - \bar{y} = b_{yx}(x - \bar{x}) = \frac{r\sigma_y}{\sigma_x} (x - \bar{x})$
Regression Coefficient (b_{xy}) of X on Y	$b_{xy} = \frac{r\sigma_x}{\sigma_y}$
Regression Line of X on Y	$x - \bar{x} = b_{xy}(y - \bar{y}) = \frac{r\sigma_x}{\sigma_y} (y - \bar{y})$
Relationship between coefficients	$b_{xy} \cdot b_{yx} = r^2$

IV. Hypothesis Testing

A. Chi-Square Test

Used to test independence or association between variables. O_i is observed value, E_i is expected value.

Concept	Formula
Chi-Square Test Statistic (χ^2_{calc})	$\chi^2_{calc} = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$
Expected Value (E_i)	$E_i = \frac{\text{Row sum} \times \text{Column sum}}{\text{Total}}$

B. Small Sample Tests (t-tests)

Used when sample size $n < 30$ and population S.D. (σ) is unknown.

Concept	Formula
One-Sample t-test	$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$
Sample Standard Deviation (S)	$S = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$
Paired t-test (Dependent)	$t = \frac{\bar{d} - d_0}{S_d/\sqrt{n}} \text{ (}\bar{d} \text{ is the mean difference)}$
Independent t-test (Two Samples)	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$